multiomicsGEP — Project Status

Supervised Bayesian Matrix Factorization for Joint Genomics + Survival Modelling

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0.1 Project Overview

This project implements a **Supervised Bayesian Matrix Factorization** model that jointly decomposes high-dimensional genomics data (gene expression, DNA methylation, etc.) and models patient survival outcomes. The core idea is to find a shared latent space that simultaneously explains genomic variation *and* predicts clinical outcomes.

The model:

- **Genomics:** Y = L F' + E, where L (n x K) are patient loadings, F (p x K) are biological factor weights, and E is Gaussian noise with feature-specific precision tau_j.
- **Survival:** Cox proportional hazards h(t_i) = h_0(t_i) exp(sum_k l_ik * beta_k), linking the same latent loadings L to time-to-event data via survival coefficients beta.

Inference: Coordinate Ascent Variational Inference (CAVI), where each parameter update reduces to an Empirical Bayes Normal Means (EBNM) problem with point-normal priors.

0.2 Repository Map

```
multiomicsGEP/
|-- CLAUDE.md                        <- Claude Code entry point (thin; defers here)
|-- PROJECT_STATUS.qmd/.pdf              <- This file (project documentation)
|-- README.md                             <- GitHub front page
|-- code/
|   |-- SupervisedMF_Context.md    <- AI/collaborator quick-reference for V2 code
|   |-- Supervised_Bayesian_MF.R    <- V1: Original CAVI implementation (reference only)
|   |-- Supervised_Bayesian_MF_V2.R  <- V2: Corrected implementation (all fixes) -- USE THIS
|    |-- update_beta.R                    <- Modular beta update functions
|   |-- update_L.R             <- Modular q(L) update functions (vector EBNM, dual-source)
|   |-- update_F.R             <- Modular q(F) update functions (pure genomics, tau cancella
|   |-- update_tau.R           <- Modular q(tau) update functions (closed-form MLE)
|   `-- multiomicsGEP_code.Rmd      <- Earlier exploratory R Markdown notebook
|-- derivations/
|    |-- qB/
|   |   |-- qBeta_update_derivation.tex  <- Full step-by-step q(beta) derivation (11 pages)
|    |    `-- qBeta_update_derivation.pdf
|    |-- qL/
|   |   |-- qL_update_derivation.tex    <- Vector EBNM, dual-source (genomics+survival)
|    |    `-- qL_update_derivation.pdf
|    |-- qF/
|   |   |-- qF_update_derivation.tex    <- tau cancellation property, pure genomics
|    |    `-- qF_update_derivation.pdf
|    |-- qTau/
|   |   |-- qTau_update_derivation.tex  <- Closed-form MLE, variance correction
|    |    `-- qTau_update_derivation.pdf
```

```
|    |-- EBMF/
|    |    `-- EBMF_Derivations*.pdf        <- Empirical Bayes MF theory notes
|    |-- MF_UpdateDerivations/
|   |   |-- MF_Derivations_UpdateAlgo_2_12_26.pdf  <- Reviewed draft -- HAS ERRORS (R1-R8)
|   |   |-- MF_Derivations_UpdateAlgo_REVISED.tex/.pdf  <- Corrected (21 pages)
|    |    `-- MF_V2_Companion.tex/.pdf    <- Math <-> code mapping (17 pages)
|    `-- SurvivalMF/
|        `-- *.pdf                        <- Survival + MF background notes
|-- results/
|   |-- run_simulation.R          <- Reproduce V2 simulation (8 figs, 11 CSVs)
|   |-- run_modular_simulation.R    <- Reproduce modular simulation (standalone)
|   |-- simulation_report.qmd/.pdf  <- V2 simulation report (Quarto, ggplot2)
|   |-- modular_sim_report.qmd/.pdf <- Modular simulation report (Quarto, per-fig commentar
|    |-- simulation_console_output.txt
|    |-- figures/
|    |    |-- full_sim/               <- 8 figure pairs from V2 simulation
|    |   `-- modular_sim/           <- 8 figure pairs from modular simulation
|    `-- tables/
|        |-- full_sim/               <- 11 CSV tables from V2 simulation
|       `-- modular_sim/            <- 11 CSV tables from modular simulation
|-- tests/
|    |-- test_helpers.R              <- Lightweight assertion framework
|    |-- test_update_beta.R          <- 24 tests for update_beta.R (9 groups)
|    |-- test_update_L.R             <- 28 tests for update_L.R (9 groups)
|    |-- test_update_F.R             <- 26 tests for update_F.R (9 groups)
|    |-- test_update_tau.R           <- 27 tests for update_tau.R (9 groups)
|    `-- run_tests.R                 <- Master runner: 105/105 tests passing
|-- demos/
|    |-- demo_update_beta.R          <- 5 demos for update_beta.R
|    |-- demo_update_L.R             <- 5 demos for update_L.R
|    |-- demo_update_F.R             <- 5 demos for update_F.R
|    `-- demo_update_tau.R           <- 5 demos for update_tau.R
|-- docs/
|    |-- Makefile                   <- `make all` renders .md -> .pdf + .html
|    |-- update_beta.md/.pdf/.html    <- Companion doc for update_beta.R
|    |-- update_L.md/.pdf/.html       <- Companion doc for update_L.R
|    |-- update_F.md/.pdf/.html       <- Companion doc for update_F.R
|    `-- update_tau.md/.pdf/.html     <- Companion doc for update_tau.R
`-- paper/
    `-- multiomicsGEP_manuscript.qmd  <- Manuscript draft (Quarto)
```

0.3 Mathematical Framework

0.3.1 Model Components

Symbol	Dimension	Meaning
Y	n x p	Observed genomics data matrix
L	n x K	Patient loading matrix (shared latent space)
F	p x K	Biological factor weight matrix
beta	K x 1	Survival coefficient vector
tau_j	p-vector	Feature-specific noise precision
(t_i, delta_i)	n-vectors	Survival time and event indicator

0.3.2 Inference Strategy

1. **Mean-field factorisation:** q(L, F, beta) = q(L) q(F) q(beta)
2. **CAVI loop:** For each factor k = 1, …, K:
    - Update q(l_k) via EBNM(x = B_L/A_L, s = 1/sqrt(A_L))
    - Update q(f_k) via EBNM(x = B_F/A_F, s = 1/sqrt(A_F))
    - Update q(beta_k) via EBNM(x = B_beta/A_beta, s = 1/sqrt(A_beta))
3. **Update tau_j** from expected squared residuals
4. **Cox Taylor expansion** linearises the non-conjugate survival likelihood into a weighted Gaussian form with working response z_i and weight W_ii

0.3.3 Key Mathematical Concepts

- **Posterior means vs second moments:** EL = E_q[l], EL2 = E_q[l^2] = Var_q(l) + mean^2. This distinction is critical for the variance inflation term in the tau update and the error-in-variables correction in the beta update.
- **EBNM sub-problems:** Each coordinate update reduces to: given pseudo-observations x with noise s, estimate a prior g and posterior q.
- **Error-in-variables:** Using E[l^2] (not l-bar^2) in the beta precision prevents overfitting to uncertain loadings.

0.4 Derivation Review Summary

A thorough review of `MF_Derivations_UpdateAlgo_2_12_26.pdf` identified **8 errors**, all corrected in `MF_Derivations_UpdateAlgo_REVISED.tex`.

> ⚠ Derivation errors in 2_12_26.pdf (R1–R8)
>
> The February 12 draft contains 8 errors listed below. Use **REVISED.pdf** for all reference.

ID	Location	Error	Correction
R1	Sec. 1A	Wrong dimension subscripts on Y, L, F, E	Fixed: Y_(nxp) = L_(nxK) F'*(Kxp)* + *E*(nxp)
R2	Eq. 22	z^{-k}*i defined with f*{jk'} instead of beta_{k'}	Fixed: uses beta_{k'} (survival coefficient)
R3	Eqs. 32-35	f^2_{jk}, beta^2_k where E_q second moments needed	Added explicit E_q notation
R4	Eqs. 47-48	EBNM inputs subscripted i (sample)	Fixed to j (feature) for F update
R5	Eq. 62	Missing exponent: beta_k should be beta^2_k	Added squared exponent
R6	Sec. 6	beta update as generic WLS; no EBNM form	Derived full A_k, B_k for EBNM
R7	Eq. 83	Sign error: log tau + tau * R-bar^2	Fixed to log tau - tau * R-bar^2
R8	Eq. 85	l^2_{jk} wrong index (j should be i)	Fixed to l^2_{ik}

0.4.1 Document Relationships

- **2_12_26.pdf** — reviewed document; contains errors R1–R8
- **REVISED.tex/.pdf** — corrected version with all 8 fixes + errata table (21 pages)
- **MF_V2_Companion.tex/.pdf** — comprehensive companion mapping math to V2 code (17 pages)

0.5 Code Versions

0.5.1 V1: `Supervised_Bayesian_MF.R`

The original implementation. Functionally correct but with several known issues:

- Orthogonalisation always ON; resets EL2/EF2 to squared means (drops posterior variance every 10 iterations)
- z_no_k computed from start-of-iteration eta only (not true Gauss-Seidel)
- No numerical floors on EBNM precision inputs

- Convergence checks only delta-L (not delta-beta)
- Only RMSE tracked (no ELBO monitoring)

0.5.2 V2: `Supervised_Bayesian_MF_V2.R`

> **Recommended implementation**
>
> Use `code/Supervised_Bayesian_MF_V2.R` for all runs. V1 is retained for reference only.

The corrected implementation with 6 algorithmic improvements:

ID	Change	Description
A1	Gauss-Seidel CAVI	z_no_k recomputed from current EL, EBeta inside k-loop
A2	Orthogonalisation flag	Behind `orthogonalize=FALSE` (default OFF)
A3	Precision floors	`pmax(..., 1e-10)` on A_L, A_F, A_Beta
A4	Dual convergence	Both mean
A5	ELBO proxy	Genomics log-likelihood tracked per iteration
A6	Taylor refresh flag	`refresh_taylor=FALSE` for optional per-k recomputation

Function signature:

```
fit_supervised_mf(Y, time, status, K=5, max_iter=100, tol=1e-5,
                  orthogonalize=FALSE, refresh_taylor=FALSE, verbose=TRUE)
# Returns: list(L, F, Beta, Beta2, Tau, history)
```

0.6 How to Run

0.6.1 Prerequisites

```
install.packages(c("survival", "ebnm"))
```

0.6.2 Running the Simulations

V2 (monolithic):

```
source("code/Supervised_Bayesian_MF_V2.R")
```

Modular (standalone, recommended for understanding the algorithm):

```
Rscript results/run_modular_simulation.R
```

Both use the same parameters (n=250, p=1000, K=5, seed=42) and produce equivalent results. The modular script sources only the four standalone update modules; it does not depend on V2.R.

0.6.3 Expected Output

- **RMSE** converges near 1.0 (the true noise SD)
- **ELBO proxy** is non-decreasing
- **Estimated beta** recovers sign pattern (+, -, +, -, 0)
- **Supervised C-index** ~0.86 on held-out tertiles
- **Censoring rate** ~30-40%

0.6.4 Compiling LaTeX Documents

```
cd derivations/MF_UpdateDerivations/
pdflatex MF_Derivations_UpdateAlgo_REVISED.tex
pdflatex MF_V2_Companion.tex
pdflatex MF_V2_Companion.tex   # second pass for cross-references

cd derivations/qB/
pdflatex qBeta_update_derivation.tex  # run 3x for stable bookmarks
```

0.6.5 Running the Test Suite

> **i** Test suite status: 105/105 passing
>
> Run after any change to a modular update script: `Rscript tests/run_tests.R`

```
Rscript tests/run_tests.R
# Expected: 105/105 tests passed (24 beta + 28 L + 26 F + 27 tau)
```

0.6.6 Running the Modular Update Demos

```
Rscript demos/demo_update_beta.R   # beta update: 5 demos
Rscript demos/demo_update_L.R      # q(L) update: 5 demos
Rscript demos/demo_update_F.R      # q(F) update: 5 demos -- see tau cancellation
```

```
Rscript demos/demo_update_tau.R    # q(tau) update: 5 demos -- see variance correctio
```

0.7 Current Status & Next Steps

0.7.1 Completed

- ☒ Derivation review: 8 errors (R1-R8) identified and corrected
- ☒ REVISED.tex/PDF: Corrected derivation document with errata table (21 pages)
- ☒ REVISED.tex expanded: Full step-by-step algebra added throughout (Taylor, L/F/beta/tau updates) with new `derivbox` gray-box environment for intermediate steps
- ☒ V2.R: Corrected R implementation with all 6 algorithmic fixes (A1-A6)
- ☒ V2_Companion.tex/PDF: Comprehensive math-to-code companion document (17 pages)
- ☒ PROJECT_STATUS + SupervisedMF_Context.md: Project documentation
- ☒ Simulation validation: V2 runs end-to-end on synthetic data (n=250, p=1000, K=5)
- ☒ results/: Full simulation outputs – 8 figures, 11 CSV tables, console log
- ☒ derivations/qB/qBeta_update_derivation.tex/.pdf: Self-contained q(beta) derivation (11 pages)
- ☒ code/update_beta.R: Modular beta update (`compute_z_no_k`, `update_beta_k`, `update_beta_all`)
- ☒ tests/: Test infrastructure + 24 tests for beta update (9 groups, all passing)
- ☒ demos/demo_update_beta.R: 5 interactive demos for beta update
- ☒ code/update_L.R: Modular q(L) update – vector EBNM, dual-source
- ☒ code/update_F.R: Modular q(F) update – pure genomics, tau cancellation
- ☒ code/update_tau.R: Modular q(tau) update – closed-form MLE
- ☒ tests/test_update_L.R: 28 tests (9 groups, all passing)
- ☒ tests/test_update_F.R: 26 tests (9 groups, tau cancellation verified)
- ☒ tests/test_update_tau.R: 27 tests (9 groups, variance correction verified)
- ☒ tests/run_tests.R: Updated master runner – 105/105 tests passing
- ☒ demos/demo_update_L.R: 5 demos (anatomy, signal recovery, genomics vs combined, error-in-variables, K=5)
- ☒ demos/demo_update_F.R: 5 demos (tau cancellation, sparse recovery, differential shrinkage, sparsity, K=5)
- ☒ demos/demo_update_tau.R: 5 demos (anatomy, known noise, variance correction, heteroscedastic, ELBO proxy)
- ☒ derivations/qL/qL_update_derivation.tex/.pdf: Self-contained q(L) derivation
- ☒ derivations/qF/qF_update_derivation.tex/.pdf: Self-contained q(F) derivation
- ☒ derivations/qTau/qTau_update_derivation.tex/.pdf: Self-contained q(tau) derivation
- ☒ docs/: Companion documentation for all 4 modular update scripts (.md + .pdf + .html)
- ☒ roxygen2 @export/@family/@seealso tags added to all 11 functions
- ☒ Targeted inline comments added to all 4 R scripts
- ☒ results/ reorganized: figures/{full_sim,modular_sim}/ and tables/{full_sim,modular_sim}/
- ☒ results/run_modular_simulation.R: Standalone CAVI script sourcing only the 4 modular modules

- ☒ results/modular_sim_report.qmd/.pdf: Quarto simulation report for modular implementation
- ☒ results/simulation_report.qmd/.pdf: Quarto simulation report for V2 implementation
- ☒ CLAUDE.md created; PROJECT_STATUS converted to .qmd for PDF rendering

0.7.2 Potential Next Steps

- ☐ **Real data application:** Apply V2 to actual genomics + survival datasets (TCGA, GEO, etc.) using the `DATA_MODE = "real"` pathway in V2.R
- ☐ **Select K:** Implement cross-validated C-index or held-out log-likelihood to choose the number of factors K objectively (currently set to 5)
- ☐ **Additional prior families:** Test `prior_family = "point_laplace"` or `"normal_scale_mixture"` in EBNM calls for different sparsity structures
- ☐ **Full ELBO:** Track the complete ELBO (survival Taylor term + all KL divergences) instead of the genomics-only proxy currently used
- ☐ **Scalability:** Profile and optimise for large p (e.g., p > 10,000) – focus on the n x p `Var_Term` and `R_k` matrix operations
- ☐ **Manuscript:** Complete `paper/multiomicsGEP_manuscript.qmd`; reference V2 results and corrected derivations
- ☐ **Bibliography:** Add `\bibliography{refs}` to REVISED.tex to resolve the `\citep{wang2022}` undefined-reference warning in the compiled PDF

0.8 Session Log

0.8.1 Session 1 (March 4–5, 2026)

Derivation review and code correction session.

1. Reviewed `MF_Derivations_UpdateAlgo_2_12_26.pdf` against the code
2. Identified 8 derivation errors (R1–R8)
3. Created `MF_Derivations_UpdateAlgo_REVISED.tex` with all corrections + errata table
4. Analysed V1 code against corrected derivations; identified 6 improvements
5. Created `Supervised_Bayesian_MF_V2.R` with all algorithmic fixes (A1–A6)
6. Created `MF_V2_Companion.tex` (math to code companion, 17 pages)
7. Created `PROJECT_STATUS.md` and `SupervisedMF_Context.md`

Key decisions:

- Orthogonalisation defaults to OFF in V2 (Point-Normal EBNM promotes sparsity/distinctness without rotation)
- Taylor refresh defaults to OFF (standard IRLS-within-VI; more stable)
- V1 preserved as-is for reference; V2 is the recommended implementation

Deliverables committed: pushed to `origin/main`

0.8.2 Session 2 (March 5, 2026)

Simulation validation, derivation expansion, and documentation session.

1. **Expanded `REVISED.tex`** from ~650 lines to 1,199 lines with full intermediate algebra throughout every update section:

   - Added new `derivbox` tcolorbox (gray) for algebra steps, distinct from `correctionbox` (red) and `ebnmbox` (blue)
   - **Taylor (Sec 3):** Full 15-step completing-the-square derivation
   - **L update (Sec 4):** Explicit residual substitution, A/B identification for genomics and survival terms separately then combined
   - **F update (Sec 5):** Full quadratic grouping to A_jk, B_jk with R4 fix
   - **beta update (Sec 6):** Variance-mean decomposition E[l^2] = Var(l) + l-bar^2 shown explicitly; error-in-variables correction A_k derived (R5, R6)
   - **tau update (Sec 7):** Full R-bar^2_ij expansion; explicit optimisation step (R7, R8)
   - Recompiled REVISED.pdf: **21 pages** (was 12)

2. **Ran V2.R simulation** (n=250, p=1000, K=5):

   - Key results: RMSE = 0.9978, C-index PCA = 0.827 vs Supervised = 0.828, PH test p = 0.258 (no violation), convergence in 45 iterations

3. **Wrote and rendered `results/simulation_report`** – 12-section comprehensive report, 23 pages via pandoc + xelatex

Deliverables committed: `5da14f3`, `e424d14` (pushed to `origin/main`)

0.8.3 Session 3 (March 10, 2026)

q(beta) derivation, modular implementation, and TDD test suite.

1. **Verified q(beta) derivation** across REVISED.tex (Sec. 6) and March 6 PDF (Sec. 6A):

   - Both agree: A_k = sum_i W_ii * E_q[l^2_ik], B_k = sum_i W_ii * z^{-k}_i * l-bar_ik
   - Confirmed March 6 PDF Eq. 68 typo is intermediate-step only; flagged as R5 in REVISED
   - Confirmed V2.R omits 1/sigma^2 prior term intentionally (EBNM handles it empirically)
   - Proved z_no_k reuse is mathematically valid
   - Confirmed: s_k = 1/sqrt(A_k) is SMALLER when A_k is larger (not larger)

2. **Created test infrastructure** (`tests/` directory):

   - `tests/test_helpers.R`: assert_near, assert_true, assert_equal, assert_length, etc.
   - `tests/run_tests.R`: master runner; exits status 1 if any tests fail

3. **TDD Red phase**: 24 failing tests first (9 groups: math identities, WLS limit, K=1 recovery, K=5 recovery, null shrinkage, error-in-variables, numerical stability, Gauss-Seidel ordering, V2 consistency)
4. **TDD Green phase**: Implemented `code/update_beta.R`:
    - `compute_z_no_k(z, EL, EBeta, k)` – partial working response
    - `update_beta_k(w, z_no_k, EL_k, EL2_k, ...)` – single-factor update
    - `update_beta_all(w, z, EL, EL2, EBeta, ...)` – full Gauss-Seidel loop
5. **Fixed 2 test logic errors** during green phase:
    - T4.1: `length()` returns integer (5L != 5); fixed to `assert_length()`
    - T6.1: s_high should be < s_low (not >); larger A_k -> smaller s_k
6. **Created `derivations/qB/qBeta_update_derivation.tex`** (11 pages):
    - tcolorbox styles: derivbox (gray), correctionbox (red), ebnmbox (blue), verifybox (green), codebox (orange)

Key decisions:

- Plain R test framework (no testthat/DESCRIPTION) – no external dependencies
- `update_beta_k` decoupled from CAVI loop; returns diagnostic fields (A, B, x, s)
- Gauss-Seidel propagation: EBeta_curr[k] updated immediately within the loop

---

0.8.4 Session 4 (March 11, 2026)

Demo for beta update + to-do setup for L/F/tau modules.

1. Created `demos/demo_update_beta.R` – 5 interactive scenarios: anatomy of A_k/B_k, signal recovery K=1, sparsity/shrinkage, error-in-variables, multi-factor K=5. Format: sep/sec/val helpers, pure narrative output (no PASS/FAIL).
2. Updated `PROJECT_STATUS.md` to add `demos/` to the repo map.
3. Updated `code/SupervisedMF_Context.md` with expanded L/F/tau next-step details.

Deliverables committed: `ad80c4e` (pushed to `origin/main`)

0.8.5 Session 5 (March 12, 2026)

Full implementation of modular updates for q(L), q(F), and q(tau).

1. **`code/update_L.R`** – Three functions:
   - `compute_R_k(Y, EL, EF, k)`: Partial residual matrix (n x p), shared with F update
   - `update_L_k(...)`: Vector EBNM – A_L[i] and B_L[i] are n-vectors (patient-specific Cox weights W_ii); two additive sources (genomics + survival)
   - `update_L_all(...)`: Gauss-Seidel loop over K factors
2. **`code/update_F.R`** – Two functions:
   - `update_F_k(...)`: Pure genomics EBNM – tau_j **cancels** in x_j = B_F[j]/A_F[j] but does NOT cancel in s_j
   - `update_F_all(...)`: Gauss-Seidel loop over K factors
3. **`code/update_tau.R`** – Three functions:
   - `compute_var_term(EL, EL2, EF, EF2)`: Variance correction (EL2 x EF2' - EL^2 x EF'^2)
   - `compute_expected_residual_sq(...)`: R2_bar = naive residual^2 + Var_Term
   - `update_tau(...)`: Closed-form MLE tau-hat_j = n / colSums(R2_bar) + ELBO proxy
4. **Tests** (9 groups each, all passing):
   - 28 tests for update_L.R (T9 verifies bit-for-bit match with V2.R lines 310–321)
   - 26 tests for update_F.R (T1.4/T1.5 verify tau cancellation numerically)
   - 27 tests for update_tau.R (T4 verifies variance correction direction)
   - Master runner: 105/105 total tests passing
5. **Demos** (5 scenarios each) and **LaTeX derivations** (compiled to PDF) for all three modules.

Key design decisions:

- `compute_R_k` shared between update_L.R and update_F.R (F sources it from update_L.R)
- tau does not appear in the survival term – F and tau updates are pure genomics
- `update_tau` has NO per-k loop and NO ebnm dependency (only base R)
- ELBO proxy computed inside `update_tau` as a convergence monitor

0.8.6 Session 6 (March 20, 2026)

Comprehensive documentation: companion docs, roxygen2, and inline comments.

1. **Created `docs/` directory** with companion documentation for all four modules:
   - `docs/update_beta.md` – Overview, math background, function reference, test + demo explanations
   - `docs/update_L.md` – Dual-source precision/signal, compute_R_k shared dependency

- `docs/update_F.md` – tau cancellation property, pure genomics
- `docs/update_tau.md` – Closed-form MLE, variance correction, ELBO proxy

2. **Created `docs/Makefile`**: `make all` renders all 4 `.md` files to PDF + HTML (Helvetica/Menlo fonts for Unicode Greek letter support).
3. **Added roxygen2 package-readiness tags** to all 11 functions across 4 R scripts:
   - `@export`, `@family` (beta_update / L_update / F_update / tau_update), `@seealso`
4. **Added targeted inline comments** (~3–5 per file) explaining *why* at non-obvious points.
5. **Verification:** 105/105 tests still passing; all 8 rendered outputs (4 PDF + 4 HTML) confirmed.

0.8.7 Session 7 (March 20, 2026)

Modular simulation report, results/ reorganization, and Quarto conversion.

1. **Reorganized `results/`**:
   - Created `figures/{full_sim,modular_sim}/` and `tables/{full_sim,modular_sim}/`
   - Moved existing V2 figures/CSVs into `full_sim/` subfolders
   - Updated `run_simulation.R` output paths accordingly
2. **Created `results/run_modular_simulation.R`** (~340 lines):
   - Completely standalone – sources only the 4 modular update modules
   - `calc_cox_taylor()` copied verbatim from V2.R; no other V2 dependency
   - Block-by-type update ordering: all L -> all F -> all beta -> tau (vs. V2's by-factor ordering; equivalent CAVI objective)
   - Results: RMSE=0.9979, C-index=0.858, PH test p=0.44, null factor beta correctly zeroed
3. **Created `results/modular_sim_report.qmd`** (863 lines):
   - Self-contained Quarto PDF; CAVI runs inline with `#| cache: true`
   - ggplot2 figures (theme_bw), kableExtra tables, Quarto callout blocks
   - Per-figure commentary (~150–300 words each) with statistical insights
4. **Created `results/simulation_report.qmd`** (786 lines):
   - Sources V2.R inline with `pdf(NULL)` + `sink()` wrappers to suppress output
5. Deleted `.md` report files; `.qmd` + `.pdf` are the canonical report artifacts.

Deliverables committed: `9a59c3e` (pushed to `origin/main`)

0.8.8 Session 8 (March 20, 2026)

Documentation housekeeping: CLAUDE.md, file role clarification, PROJECT_STATUS to Quarto.

1. **Created `CLAUDE.md`** – thin Claude Code entry point (key instructions + quick-reference table); defers to `PROJECT_STATUS` for full context.
2. **Discussed file role separation:**
   - `README.md` = GitHub front page (overview, quickstart, brief math)
   - `PROJECT_STATUS.qmd/.pdf` = full internal working document (session log, derivation review, algorithm design decisions, next steps)
   - `CLAUDE.md` = lean Claude Code instructions pointing to PROJECT_STATUS
3. **Converted `PROJECT_STATUS.md` to `PROJECT_STATUS.qmd`** for PDF rendering:
   - Added Quarto YAML header (Helvetica Neue / Menlo, TOC, numbered sections)
   - Added callout blocks for key notices (derivation errors warning, V2 recommended tip, test suite status note)
   - Deleted `PROJECT_STATUS.md`; `.qmd` + `.pdf` are now canonical
4. Updated `MEMORY.md` documentation convention to reflect the three-file structure.

Deliverables committed: pushed to `origin/main`